ZZ-M-91e
February 28, 1963

SUPERSEDING
Int. Fed. Spec. ZZ-M-0091d(Army-GL)
January 23, 1963 and
Fed. Spec. ZZ-M-91c
November 3, 1960

FEDERAL SPECIFICATION

MATTRESS, BED, LATEX FOAM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers latex foam bed mattresses.

1.2 Classification.

1.2.1 *Sizes.* Mattresses covered by this specification shall be of the sizes as specified herein (see table II and 6.1).

Size 1.

Size 2.

Size 3.

Size 4.

Size 5.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

V-F-106—Fasteners; Slide, Interlocking.

V-T-276—Thread, Cotton.

CCC-C-436—Cloth, Cotton, Ticking, Twill.

CCC-T-191—Textile Test Methods.

DDD-L-20—Label; For Clothing, Equipage and Tentage (General Use).

DDD-T-0086—Tape, Textile, Cotton, General Purpose (Unbleached, Bleached or Dyed).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 601—Rubber; Sampling and Testing.

Fed. Std. No. 751—Stitches; Seams and Stitching.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Washington.)

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Military Specification:

MIL-M-3996—Mattresses (Except Pneumatic Type), Packaging and Packing of.

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Latex-foam insert.* The latex-foam insert shall be made from synthetic-latex foam, natural-latex foam, or a mixture of natural and synthetic-latex foam. The cured foam shall consist of interconnecting cells reasonably uniform in size and free from large internal voids perceptible on the exterior of the mattress by sight or touch. The foam shall be free from unrepaired surface voids (other than coring) 1/2 inch or larger in depth and 1/2 inch or larger in width. The cured foam shall conform to the physical requirements specified in table I and 3.1.1.1 when tested as specified in 4.3.1.1.

TABLE I.—*Physical requirements of latex foam*

Characteristic	Requirement
Permanent set (maximum percent)	20
Resistance to flexing (250,000 cycles)	Note 1
Permanent set after flexing (maximum percent)	5
Compression after accelerated aging (maximum percent change)	20
Static fatigue	Note 2
Cement bond	Note 3
Odor	Note 4

[1] The latex foam shall show no breakdown of the physical structure.

[2] The latex foam shall show no cracking at the folded edge.

[3] The bond shall be stronger than the adjoining foam when separated by hand.

[4] The foam shall be free from any objectionable odor.

3.1.1.1 *Compression.* The average compression (pounds to compress 25 percent of total thickness) shall be 28-1/2 pounds ± 2 pounds. No individual specimen shall have an average compression less than 22 pounds nor more than 35 pounds, when tested as specified in 4.3.1.1.

3.1.2 *Ticking.*

3.1.2.1 *For sewn-on cover.* The ticking for sewn-on covers shall be type I of CCC-C-436.

3.1.2.2 *For removable cover.* The ticking for removable cover shall be type I of CCC-C-436 except that the ticking shall be preshrunk and the residual shrinkage shall not exceed 2.0 percent in either the warp or filling direction when tested as specified in 4.3.1.1.

3.1.3 *Thread, cotton.* The thread shall conform to type I Al, three ply, ticket No. 30 of V-T-276.

3.1.4 *Tapes.*

3.1.4.1 *Edging tape.* The tape used to edge the top and bottom perimeter of the latex foam insert shall be unbleached cotton sheeting conforming to the following properties when tested as specified in 4.3.1.1:

	Minimum
Weight, ounces per square yard	3.09
Yarns per inch—	
Warp	48
Filling	44
Breaking strength, pounds—	
Warp	45
Filling	32

In lieu of the above, a pressure-sensitive cotton tape having a minimum texture of 80 X 78 may be used (6.4).

3.1.4.2 *Closure tape.* The closure tape sewed to the top and bottom perimeter of the

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sewn-on cover shall be 5/8 inch wide, unbleached, cotton tape conforming to type I, class 1 of DDD-T-86.

3.1.5 *Slide fastener.* The slide fastener shall conform to type I, style 8, size LM, short tab pull, of V-F-106. The slide fastener chains shall be fabricated of brass or aluminum. The slide fastener shall be long enough to extend down one side and one end of the cover (see 3.2.5.2). The color of the slide fastener tape shall be neutral.

3.2 Construction.

3.2.1 *Structure.* The latex foam insert shall be constructed in a single piece or not more than two cemented crosscut sections, except that one of the following additional constructions are permissible for size adjustment.

(*a*) *For undersized latex-foam insert (width or length).* End or side walls may be added to adjust for size, not to exceed 1-1/2 inch on any one wall.

(*b*) *For oversized latex-foam inserts (width or length).* Cut off material up to but not including the first row of major core holes, remove excess material and cement edge strip to the latex-foam insert. The total length of such edging shall not exceed the combined length and width of the latex-foam insert.

(*c*) *For latex-foam inserts of inconsistent thickness.* The rough surface shall be made flat and a matching surface 1/4 to 3/4 inch thick and of the same type and density material as the unit shall be cemented to the mattress to provide the correct thickness. The cement used to adhere the various sections of the latex-foam insert shall form a bond stronger than the adjoining foam. Cemented seams shall be sufficiently soft and similar to the rest of the insert so that they may not be felt through the ticking cover with the flat of the hand.

3.2.2 *Coring.* Latex-foam inserts may have either a single zone core arrangement or a tri-zone core arrangement with the center zone having less cored out volume than the end zones. Core size, core-hole depth, and arrangement of core holes shall be at the option of the manufacturer subject to the following provisions:

(*a*) Core holes shall be not closer to the edge than their diameter.

(*b*) No core holes shall be larger than 1-1/2 inches in diameter.

(*c*) If tri-zone coring is employed, the middle (firmer) zone of core shall occupy at least one third of the total latex-foam insert area.

(*d*) The core holes may be on one surface or both surfaces of the latex-foam insert.

3.2.3 *Corners.* The corners of size 1 and size 2 latex-foam insert shall be rounded with a radius 4 inches maximum and 2 inches minimum. The corners of all other sizes shall be square or shall be rounded with a radius not exceeding 2 inches.

3.2.4 *Edging.*

3.2.4.1 *Mattress with sewn-on cover.* The top and bottom perimeter of the latex-foam insert shall have cotton edging tape (specified in 3.1.4.1) cemented to the side border. The tape shall be at least one inch wide with not less than 3/4 inch adhered to the latex-foam insert and the balance freely extended to permit it to be sewn to the top and bottom panels.

3.2.4.2 *Mattress with removable cover.* The top perimeter of the latex-foam insert shall be reinforced on the sides and top, and the bottom perimeter of the latex-foam insert shall be reinforced on the sides and bottom with the edging tape specified in 3.1.4.1 The tape shall be 2 inches ± 1/8 inch wide, with a minimum of 3/4 inch adhered

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to either surface and the remainder to the other surface.

3.2.4.3 *Adhesion of edging tape to latex foam.* The bond of the edging tape to the latex-foam insert shall be stronger than the adjoining foam, when the tape is stripped or peeled by hand. Tearing of the foam shall be regarded as sufficient evidence of the strength of the bond.

3.2.5 *Cover.* Mattresses shall be furnished with either a sewn-on cover or removable cover as specified (6.2).

3.2.5.1 *Sewn-on cover.* Each latex-foam insert shall be covered with the ticking specified in 3.1.2.1. The ticking used to construct the top panels, bottom panels, and border shall be one piece with stripes running lengthwise. The top and bottom panel shall have corners shaped to the contour of the latex-foam insert. The top panel, bottom panel, and border shall be hemmed with a 1/4-inch hem using stitch type 401, stitching type EFa-1 and 6 to 8 stitches per inch. The hemmed size of the panels shall be 1/2 inch larger in each direction than the size of the mattress. The border shall be stretched and shall be in one piece. After stretching, the border shall be seamed in the center of the foot end to the exact peripheral size of the mattress using stitch type 301, seam type SSa-1 with 10 to 12 stitches per inch. The border shall then be placed around the edge of the latex-foam insert. The top panel shall be sewn to the border with the edge tape between them and with the closure tape specified in 3.1.4.2 around all three pieces using stitch type 301, seam type BSa-1, with 10 to 12 stitches per inch. The bottom panel shall be sewn in the same manner and there shall be no looseness or puckering in the top and bottom panels of the border.

3.2.5.2 *Removable cover.* Removable covers shall be constructed from the ticking specified in 3.1.2.2. Each top and bottom panel shall be constructed of one piece of ticking with corners shaped to the contour of the latex foam insert they are intended to cover and with the stripes running lengthwise. Each panel shall be cut 3/4 inch longer and 3/4 inch wider than the finished mattress dimension to provide a 3/8-inch seam allowance on all sides. The border shall be made in two sections with the stripes running lengthwise. One section shall hold the slide fastener and the other section shall be a solid-piece construction. The slide fastener shall be positioned in the center of the border and shall run in a lengthwise direction. The slide fastener shall be sewn to the border using stitch type 301, seam type SSs-2, with 10 to 12 stitches per inch and shall be so positioned that the slide fastener is not exposed on the surface. The two sections of the border shall be joined to form a continuous piece using stitch type 301, seam type SSa-1, with 10 to 12 stitches per inch. To permit a 3/8-inch seam allowance, the length shall be 3/4 inch longer than the perimeter; and the width 3/4 inch wider than the edge thickness of the mattresses intended to be covered. The border shall be sewn to the top and bottom panels using seam type SSa-2, produced with a combination of stitch types 301 and 503, or 504. The edges of the SSa-2 seam shall be placed on the inside of the cover. When the border is sewn to the top and bottom panels, the slide fastener shall be positioned to include one side and one end of the cover and shall be not more than one inch from either corner.

3.2.5.3 *Stitches, seams and stitching.* Stitches, seams and stitching shall conform to Fed. Std. No. 751. Ends of all seams not caught in other seams or stitching shall be securely backstitched 7/8 to 1 inch. Skipped stitches or thread breaks (301 or 401 stitch types) shall be repaired using 301 stitch type.

3.3 Size. The sizes of the covered mattresses shall conform to dimensions specified in table II (1.2.1 and 6.2).

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TABLE II.—*Sizes of covered mattresses*

Dimensions	Size					Tolerance
	1	2	3	4	5	
Inches						*Inches*
Length	75	75	79	76	76	±1/2
Width	53	38	36	36	30	±1/2
Thickness at edge*	4	4	4	4	4	±3/16

* At the manufacturer's option, the mattress may be furnished with a 1/2 ± 1/16-inch crown.

Notes: (Not applicable for covered mattresses.)

1. When sizes 1 and 2 uncovered latex-foam inserts are procured separately, the insert shall be 1 inch longer, 1 inch wider and the same thickness as specified in table II. When sizes 3, 4, and 5 uncovered latex-foam inserts are procured separately, the inserts shall be one inch longer and the same width and thickness as specified in table II. Tolerance for length shall be plus 1-1/8 inches, minus 7/8 inch; for width plus 3/4 inch, minus 1/2 inch; and thickness plus or minus 3/16 inch.

2. When removable covers are procured separately, they shall have the same dimension and tolerances specified in table II.

3.4 **Marking.**

3.4.1 *Latex-foam insert.* Each latex-foam insert shall be identified with the manufacturer's name or trademark. The identification shall either be molded into the insert or shall be stamped on the insert with either dark blue or black ink. When the identification is stamped on the insert, it shall be legible after testing as specified in 4.4.4.

3.4.2 *Mattress cover.* Each ticking cover shall have an identification label conforming to type I, class 5 of DDD-L-20. It shall be caught in the box edge seam at the center of one end of the box edge panel.

3.4.3 *State legal requirements.* Additional markings as required by state health regulations, shall be affixed as directed by the laws of the state in which the mattresses are manufactured.

3.5 **Workmanship.** The finished mattress shall conform to the quality and grade of product established by this specification, and the occurrence of defects shall not exceed the specified acceptable quality levels.

4. SAMPLING INSPECTION AND TEST PROCEDURES

4.1 **Inspection responsibility.** The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Preacceptance inspection.**

4.2.1 *Preproduction inspection.* When required, inspection and testing of the preproduction sample shall be made of a completed mattress for all provisions of this specification applicable to end product examinations and tests.

4.3 **Inspection for acceptance.** Except when otherwise specified by Civil Government Agencies, inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

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TABLE III.—*Instructions for testing components*

Component lot expressed in terms of	CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per individual unit	Results Reported As	
		Requirement	Test Method	Individ. Unit	Lot Aver.		Pass or Fail	Numerically to Nearest
Latex-foam insert (1 unit)	Permanent set[1]	Table I	Method 12131 Fed. Std. No. 601	X	—	1		1 percent
	Resistance to flexing	Table I	Method 12111 Fed. Std. No. 601	X	—	1	X	
	Permanent set after flexing[2]	Table I	Method 12111 Fed. Std. No. 601	X	—	1		1 percent
	Compression after accelerated aging[3]	Table I	Par. 4.4.1 & Method 12121 Fed. Std. No. 601	X	—	1		1 percent
	Static fatigue	Table I	Par. 4.4.2	X	—	1	X	
	Cement bond of tape	3.2.4.3	Std. Comm'l.	X	—	1	X	
	Cement bond of insert (when applicable)	Table I	Std. Comm'l.	X	—	1	X	
	Odor	Table I	Std. Comm'l.	X	—	1	X	
	Compression[3]	3.1.1.1	Method 12121 Fed. Std. No. 601	—	X	2		Average of 2 determinations to nearest 0.5 lb.
	Color fastness of marking	3.4.1	Par. 4.4.4.	X	—	1	X	

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TABLE III.—*Instructions for testing compo nents (continued)*

Component lot expressed in terms of	CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per individual unit	Results Reported As	
		Requirement	Test Method	Individ. Unit	Lot Aver.		Pass or Fail	Numerically to Nearest
Ticking (2 yards)	Shrinkage Warp Filling	3.1.2.2	Method 5550 CCC-T-191	X	—	3 (each direction)		Average of 3 determinations to nearest 0.1 percent
Edging tape (1 yard)	Texture Warp Filling	3.1.4.1	Method 5050 CCC-T-191	X	—	5 (each direction)		Average of 5 determinations to nearest whole number
	Weight	3.1.4.1	Method 5041 CCC-T-191	X	—	5		Average of 5 determinations to nearest 0.1 oz.
	Breaking strength	3.1.4.1	Method 5100 CCC-T-191	X	—	5 (each direction)		Average of 5 determinations to nearest 1.0 lb.

[1] Except that specimen shall be exposed for a period of 22 hours in lieu of 46 hours.

[2] Except that the test specimen shall be in full thickness with minimum area dimension greater than the thickness.

[3] Except that two test specimens shall be taken from each test unit. One specimen shall be cut from the center and one from the end of the latex-foam insert but not closer than 4 inches from any edge. The test may be conducted on the whole latex-form insert instead of cut out specimens for initial compression. When the whole latex-foam insert is used, tests shall be made at the center and 12 inches from one end of the latex-foam insert. No measurement shall be made closer than 4 inches from any edge.

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4.3.1 *Sampling for inspection.* Except when otherwise specified by Civil Government Agencies, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot shall consist of all end items of the same size submitted for acceptance and delivery at one time.

4.3.1.1 *Inspection of components.* In addition to the quality assurance provisions of the subsidiary specifications and drawings, inspection shall be performed on components and materials listed in table III herein for the characteristics shown. The sample unit for testing of latex foam inserts shall be one complete insert. The sample unit for testing the ticking shall be 2 yards, full width, and the sample unit for testing the tape shall be 1 yard. The inspection level shall be L1 of MIL-STD-105. The acceptable quality level for each characteristic wherein the requirement is applicable to the individual unit shall be 6.5 defects per 100 units. For characteristics wherein the requirement is applicable to the sample average there shall be no evidence of failure to meet the specified requirement.

4.3.1.2 *In-process inspection.* Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operation or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements:

Requirement operation or assembly	*Requirement paragraph*
Covers for mattresses sewn to the top and bottom edging tape	3.2.5.1

4.3.2 *Intermediate inspection.*

4.3.2.1 *Visual examination of the intermediate product.* The lot size shall be expressed in units of completely fabricated latex-foam inserts. The inspection levels for purposes of determining the sample size and the acceptable quality levels (AQL's) are specified in 4.3.2.1.3. The defects found during the examinations shall be classified in accordance with 4.3.2.1.1 and 4.3.2.1.2. The sample unit for these examinations shall be one completely fabricated latex-foam insert prior to insertion into the case.

4.3.2.1.1 *Examination of latex foam insert for defects in material, construction, workmanship and marking.*

Examine	*Defect*
Material	Not as specified.
	Has objectionable odor.
	Voids (uncored surfaces).
	(1) Any void 1/2 inch or larger in depth and 1/2 inch or larger in width.
	(2) Any voids on an edge or corner over 1 inch in length or projecting more than 3/8 inch into a plane.
	(3) Any large internal void perceptible on the exterior by sight or touch.
	Loose skin, area on top or bottom surfaces greater than 6 square inches.
	More than 2 loose skin areas, regardless of size, on top and bottom surface, per mattress.
Construction and workmanship	Three or more cemented sections.
	Cemented seams not soft and similar in compression to the insert (should be felt through ticking cover with flat of hand).
	Cemented seam not secure.

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Examine	*Defect*
Construction and workmanship (cont'd)	Edging tape not type specified.
	Any taped edge showing a lack of adhesion completely across the tape, i.e., extending from one side edge of the tape to the opposite side edge of the tape.
	Any taped edges showing lack of adhesion extending 2 inches or more along one edge.
	Any top edge or bottom edge of the latex-foam insert not taped where required.
	Less than 3/4 inch of the width of the tape adhered to latex-foam surface.
	Surface tear (material missing) 3/8 inch deep or more.
	Repaired surface not level or not neatly accomplished.
	Uncemented tears (a tear shall be considered as such when it extends through the entire wall separating core holes).
	(1) More than 2, up to and including 3/8 inch in depth, in any 2-foot square area.
	(2) More than 1, over 3/8 inch in depth but not exceeding 1/2 inch in depth, in any 1-foot square area.
	(3) More than 3 total, extending more than 1/2 inch in depth.
	Thickness adjustment:
	(1) Core holes not matched.
	(2) Repair depth under 1/4 inch or over 3/4 inch.
	Length or width adjustment more than 1-1/2 inches wide.
Measurements (as applicable)	Core holes located less distance from the edge than their diameters.
Measurements (as applicable) (cont'd)	Length exceeds size specified by more than 1-1/8 inches.
	Width exceeds size specified by more than 3/4 inch.
	Length less than size specified by more than 7/8 inch.
	Width less than size specified by more than 1/2 inch.
	Thickness more or less than 3/16 inch from thickness specified.
	Corners of size 1 or 2 latex-foam insert rounded with more than a 4-inch radius.
	Corners of size 1 or 2 latex-foam insert rounded with less than a 2-inch radius.
	Corners of size 3, 4 or 5 latex foam insert rounded with more than a 2-inch radius.
	Core holes over 1-1/2 inches in diameter.

4.3.2.1.3 *Inspection levels and acceptable quality levels (AQL's)* The inspection levels and AQL's expressed in defects per hundred units shall be as follows:

For examination in	*Inspection levels*	*AQL*
4.3.2.1.1	I	4.0
4.3.2.1.2	L 6	4.0

4.3.3 *Inspection of the end item.*

4.3.3.1 *Visual examination.* The lot size shall be expressed in units of completely fabricated mattresses. The inspection levels for determining the sample size and the ac-

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ceptable quality levels (AQL's) are specified in 4.3.3.1.4. The defects found during the examinations shall be classified in accordance with 4.3.3.1.1 and 4.3.3.1.2. The sample unit for examinations in 4.3.3.1.1 and 4.3.3.1.2 shall be one completely fabricated mattress.

4.3.3.1.1 *Examination of mattresses for defects in material, construction, workmanship, and marking.*

Examine	Defect	Classification	
		Major	Minor
Material (ticking)	Abrasion marks, clearly noticeable .		X
	Broken or missing yarns:		
	—three or more contiguous regardless of length	X	
	—two contiguous 2 inches or more in length .	X	
	—single, 2 inches or more missing		X
	Crease, hard, permanently embedded	X	
	Hole, cut, or tear ..	X	
	Smash	X	
	Slub, slough off, or coarse yarn 1/8 inch or more in thickness		X
	Thin or open place .	X	
	Floats or skips:		
	—multiple 1/2 inch or more in combined warp and filling direction	X	
	—multiple less than 1/2 inch in combined warp and filling directions, or single, 1/2 inch or more in length		X
	Light place or fine filling bar, clearly noticeable	X	
	Spot, stain, or streak clearly noticeable when viewed at a distance of 6 feet .		X
Slide fastener (mattress with removable cover)	Cannot open or close chain the full distance	X	
	Slider binds, i.e., does not operate freely, smoothly, or positively	X	
	Any scoop improperly mated or does not mesh with its corresponding scoops when chain is in closed position	X	
	Slider pull becomes detached during operation	X	
	Slider disengages from either row of scoops when chain is in closed position	X	
	Finish of any metal component missing where required ..	X	
	Type or color of finish not as specified		X
	Corrosion or rust ..	X	

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Examine	Defect	Classification	
		Major	Minor
Slide fastener (mattress with removable cover) (cont'd)	Any metal components fractured, split, or broken ...	X	
	Sharp burr, splinter, sharp fin, or sharp edge which may injure the user	X	
Slide fastener tape and binding tape	Tape not as specified	X	
	Cut, hole, tear in the tape or bead ..	X	
	Not positioned as specified	X	
	Color of tape other than specified		X
Construction and workmanship	Cover loose or puckered to an extent adversely affecting appearance		X
	Cover other than type specified	X	
	Stripes of ticking cover do not run in lengthwise direction		X
	Any cover face fabricated of more than one piece of ticking		X
Seams and stitching	Stitch type not as specified		X
	Seam type not as specified		X
	Hem not as specified		X
	More or less stitches per inch than specified		X
Seams and stitching (cont'd)	Any seam open more than 1 inch	X	
	Any seam open up to and including 1 inch		X
	Note: A seam shall be classified as open when one or more stitches joining a seam are broken or when 2 or more continuous skipped stitches or run-offs occur.		
	Seam pleated or badly twisted		X
	Loose tension resulting in a loose seam		X
	Tight tension (stitches break when normal strain is applied to stitching)		X
	Not unbleached cotton thread		X
	Needle chew	X	
Identification labels	Missing, incorrect, text incomplete, or illegible		X

4.3.3.1.2 *Examination of mattress (with cover) for defects in dimension.* Any dimension exceeding the specified tolerance shall be classified as a defect.

4.3.3.1.3 *Examination of preparation for delivery.* An examination shall be made to determine that markings, materials, workmanship, contents and weights comply with the requirements of section 5 of this specification. The sample unit shall be one ship-

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ping container, fully packed, selected just prior to the closing operation. Closed shipping containers shall be examined for closure defects. The lot size shall be the number of shipping containers in the inspection lot.

Examine	*Defect*
Markings (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Materials	Any non-conforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components: such as incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping or stapling; bulged or distorted containers.
Contents	Number per container is more or less than required.
Weights	Gross or net weight exceeds requirements.

4.3.3.1.4 *Inspection levels and acceptable quality levels (AQL's).* The inspection levels and AQL's, expressed in defects per 100 units, shall be as follows:

For examination in	*Inspection level*	*AQL*	
		Major	*Total*
4.3.3.1.1	I	2.5	6.5
4.3.3.1.2	L 6	—	4.0
4.3.3.1.3	L 4	—	4.0

4.4 Test methods.

4.4.1 *Accelerated aging.* A specimen of latex foam at least 12 inches by 12 inches in full thickness, shall be subjected to the air-heat test specified in method 7221 of Fed. Test method Std. No. 601, except that the test shall be run for 22 hours at 212° ± 2° F.

4.4.2 *Static fatigue.* A specimen of latex foam 4 inches by 24 inches shall be bent parallel to the 4 inch dimension to an angle of 180° between two compression plates for 22 hours at 158° ± 2° F. The heat exposure shall be carried out in the apparatus described in method 7221 of Fed. Test Method Std. No. 601. The opening between the two plates should be equal to twice the thickness of the unfolded specimen. The folded edge of the specimen should not extend beyond the edges of the compression plate. When cored on one side, the cored side shall be folded inward.

4.4.3 *Physical dimensions.*

4.4.3.1 *Length.* The average length shall be determined in accordance with method 12011 of Fed. Test Method Std. No. 601, except that only three measurements shall be made on a test unit.

4.4.3.2 *Width.* The average width shall be determined in accordance with method 12021 of Fed. Test Method Std. No. 601, except that only three measurements shall be made on a test unit.

4.4.3.3 *Thickness.* Thickness measurements shall be taken as the average of three measurements. Tests shall be made with any measuring device graduated 1/32 inch and capable of accurate measurement without deforming the sample.

4.4.4 *Marking.* (Not applicable when the marking is molded in the insert.) A section of latex foam with the manufacturer's name or trademark marked thereon shall be immersed for one hour in water having a temperature of 80° ± 2°F. After this treatment, the wet marking shall be rubbed with a finger and observed for legibility.

5. PREPARATION FOR DELIVERY

Fed. Std. No. 102 shall be referred to for definitions and application of the various lev-

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els of packaging and packing for civil agency procurements.

5.1 Packaging. Packaging shall be level A or C as specified (6.2).

5.1.1 *Levels A and C.* Mattresses shall be packaged in accordance with the applicable requirements of MIL-M-3996.

5.1.2 For civil agency procurements, where level B packaging is specified in the contract or order, the requirement in 5.1.1 (level A) shall apply.

5.2 Packing. Packing shall be level A, B, or C as specified (6.2).

5.2.1 *Levels A, B, and C.* Mattresses shall be packed in accordance with the applicable requirements of MIL-M-3996.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.1.1 *Special marking.* Shipping containers shall be marked on two side faces with the following:

"STOW FLAT—DO NOT STOW ON END"

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-M-3996.

6. NOTES

6.1 Intended use. Mattresses covered by this specification are intended for use in hospitals and government establishments in the continental U. S. and overseas.

6.2 Ordering data. Purchasers should exercise any desired option offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Size required (1.2).

(*c*) Type of cover required (3.2.5).

(*d*) Selection of applicable levels of packaging and packing (5.1 and 5.2).

(*e*) Inspection responsibility, methods and tests (section 4).

6.3 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Mattress

In cars 40 feet 0 inches or less in length, inside measurements — 10,000; 12,000; 15,000; 18,000; 22,000; and 30,000 pounds.

In cars exceeding 40 feet 9 inches in length, inside measurements — 16,000; 18,000; 21,000; 24,000 30,000 and 40,000 pounds.

Motor:

Truckload minimum weights - 10,000; 12,000; 15,000; 18,000; 22,000 and 30,000 pounds subject to Rule 115, National Motor Freight Classification.

6.4 Pressure-sensitive tape. Pressure-sensitive cotton tape manufactured by The Seamless Rubber Company, New Haven, Connecticut, has been found suitable for this purpose.

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